

NEUROMORPHIC COMPUTING

When Computers Begin to Think Like the Human Brain

A New Era of Brain-Inspired Computing

Imagine a computer that can recognize patterns, learn from experience, and make decisions while consuming less power than a household light bulb. This futuristic idea is becoming reality through Neuromorphic Computing, a revolutionary field that designs computer systems by imitating the structure and working principles of the human brain.

The concept was first introduced by **Professor Carver Mead in the 1980s**. Instead of building faster processors alone, his vision was to create computers that process information the way biological neurons communicate—efficiently, intelligently, and with minimal energy.

Why Do We Need Neuromorphic Computing?

Modern computers have achieved incredible processing speeds, yet they consume enormous amounts of power. For example, the human brain operates on only about 20 watts of power while performing an estimated 10^{16} – 10^{18} operations per second

making it one of the most energy-efficient computing systems known. In comparison, some of the world's fastest supercomputers require megawatts of electricity to perform similar large-scale computations. This remarkable difference has inspired scientists to design computers that work more like the brain.

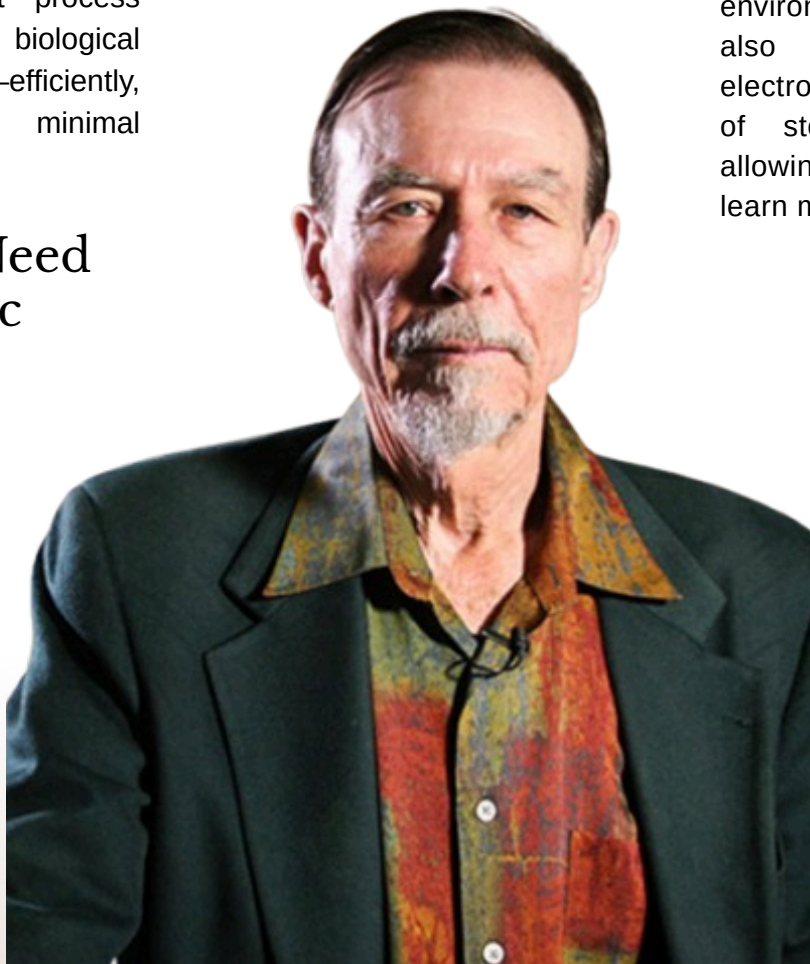
How Does It Work?

Unlike traditional computers that continuously transfer data between the processor and memory, neuromorphic systems combine processing and memory much like neurons and synapses in the human brain. These systems rely on Spiking Neural Networks (SNNs)

where artificial neurons communicate only when necessary by sending tiny electrical pulses called spikes. Because computation occurs only when an event happens, neuromorphic chips consume far less energy while responding quickly to changing environments. Many designs also incorporate memristors, electronic components capable of storing previous states, allowing systems to adapt and learn more naturally.



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"THE FUTURE OF COMPUTING IS NOT MEASURED ONLY BY SPEED, BUT BY HOW EFFICIENTLY MACHINES CAN THINK, LEARN, AND ADAPT."